

Craftax runs compared: achievements and what unlocks them

7 finished runs, 1.1M env steps each, size50m, one training seed per run. Scored with tools/death_eval.py on the same fixed evaluation worlds. 2026-09-28.

Every number is the mean episode-final achievement count over the evaluation episodes, and each unlock rate is the share of episodes in which that achievement fired. Every run was scored on the same fixed evaluation worlds. Re-scoring the old 5.80 record and the 2.30 run with today's code reproduced both exactly (same totals, per-achievement rates, death counts and lifespans), so the differences below come from the trained policies, not from how they were measured.

Two things look wrong. First, the newest run has the highest total (5.93) but never makes a pickaxe: 0 of 30 episodes, against 9 of 30 for the old record, so it never reaches stone. It makes a wood sword in 60% of episodes instead. The total went up while the tech tree got shallower.

Second, every map run unlocks WAKE_UP (sleep, then wake) in 83-98% of episodes; vanilla never does. WAKE_UP is the map runs' whole lead over vanilla: without it the fixed map model scores 4.20 against vanilla's 4.44. Only the potential shaping adds tech-tree progress on top.

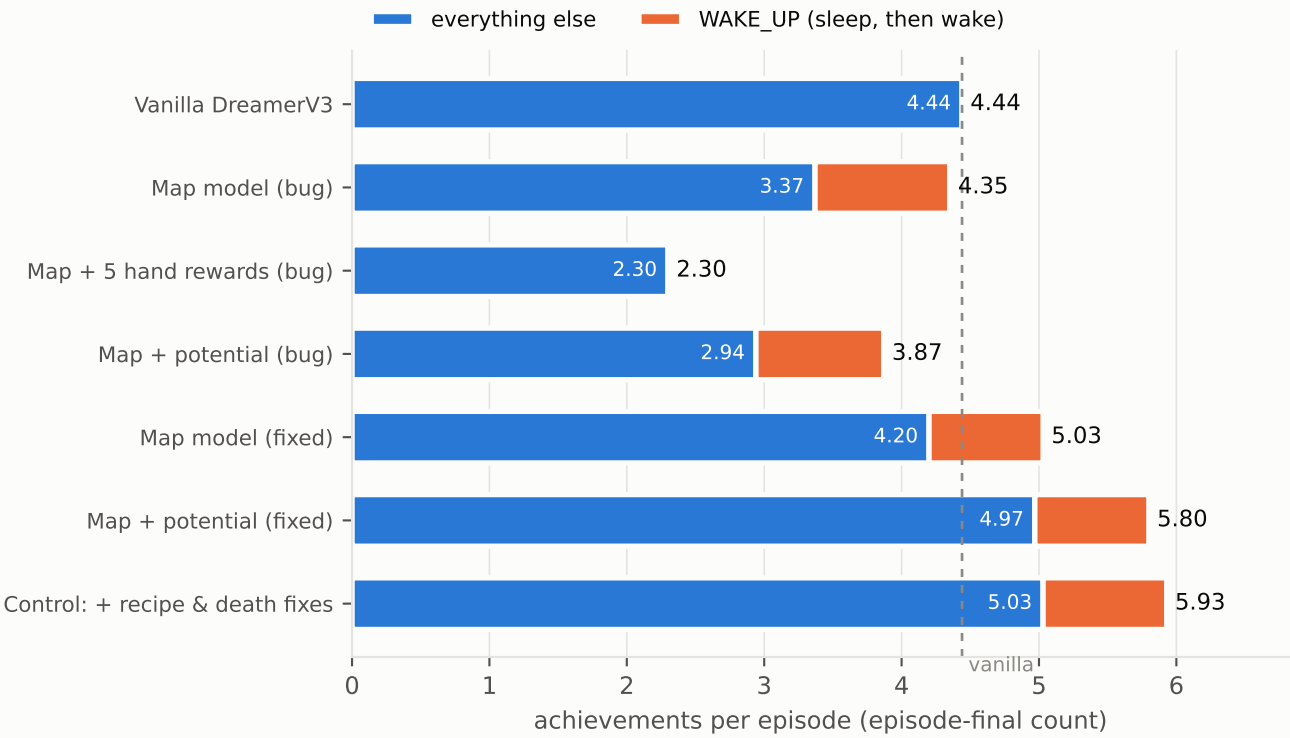
Survival never moved: every run lives 262-294 steps, inside the random-policy band (261-284), and thirst is the main killer. The non-cheating run (honest_map) is still training after an out-of-memory restart and will be added.

run	eps	ach.	w/o wake	life	table	pickaxe	sword	stone	drink
Vanilla DreamerV3	25	4.44	4.44	262	64%	24%	0%	12%	80%
Map model (bug)	60	4.35	3.37	294	35%	0%	0%	0%	43%
Map + 5 hand rewards (bug)	50	2.30	2.30	263	0%	0%	0%	0%	44%
Map + potential (bug)	30	3.87	2.94	277	13%	0%	0%	0%	37%
Map model (fixed)	30	5.03	4.20	264	63%	7%	0%	0%	70%
Map + potential (fixed)	30	5.80	4.97	278	90%	30%	10%	20%	60%
Control: + recipe & death fixes	30	5.93	5.03	268	83%	0%	60%	0%	67%

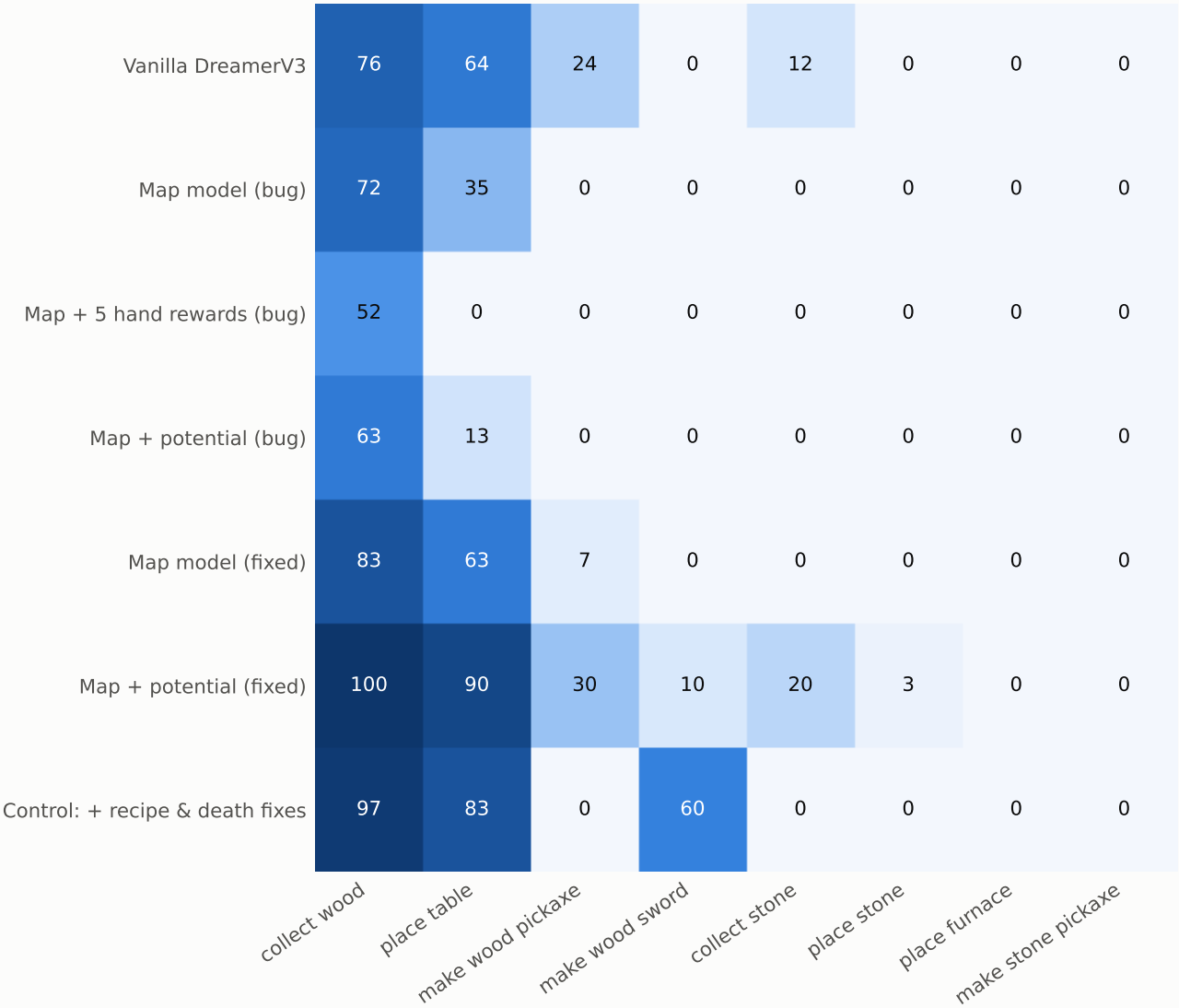
eps = evaluation episodes. ach. = mean episode-final achievements; w/o wake = the same minus WAKE_UP. life = mean lifespan in steps. Rates are % of episodes. 'bug' = imag_shift True, which corrupted the policy gradient (fixed 2026-09-20). map_survive was re-evaluated on 2026-09-28 and reproduced its original 2.30. One training seed per run, so the run-to-run spread is unknown. 'Map + potential (fixed)' is the old 5.80 record; 'Control' adds the recipe and death fixes with the privileged map.

Most of the gain over vanilla is sleep

Mean episode-final achievements per run, split into WAKE_UP and the rest; unlock rate of the early tech-tree rungs



Unlock rate (% of episodes), early tech tree in dependency order



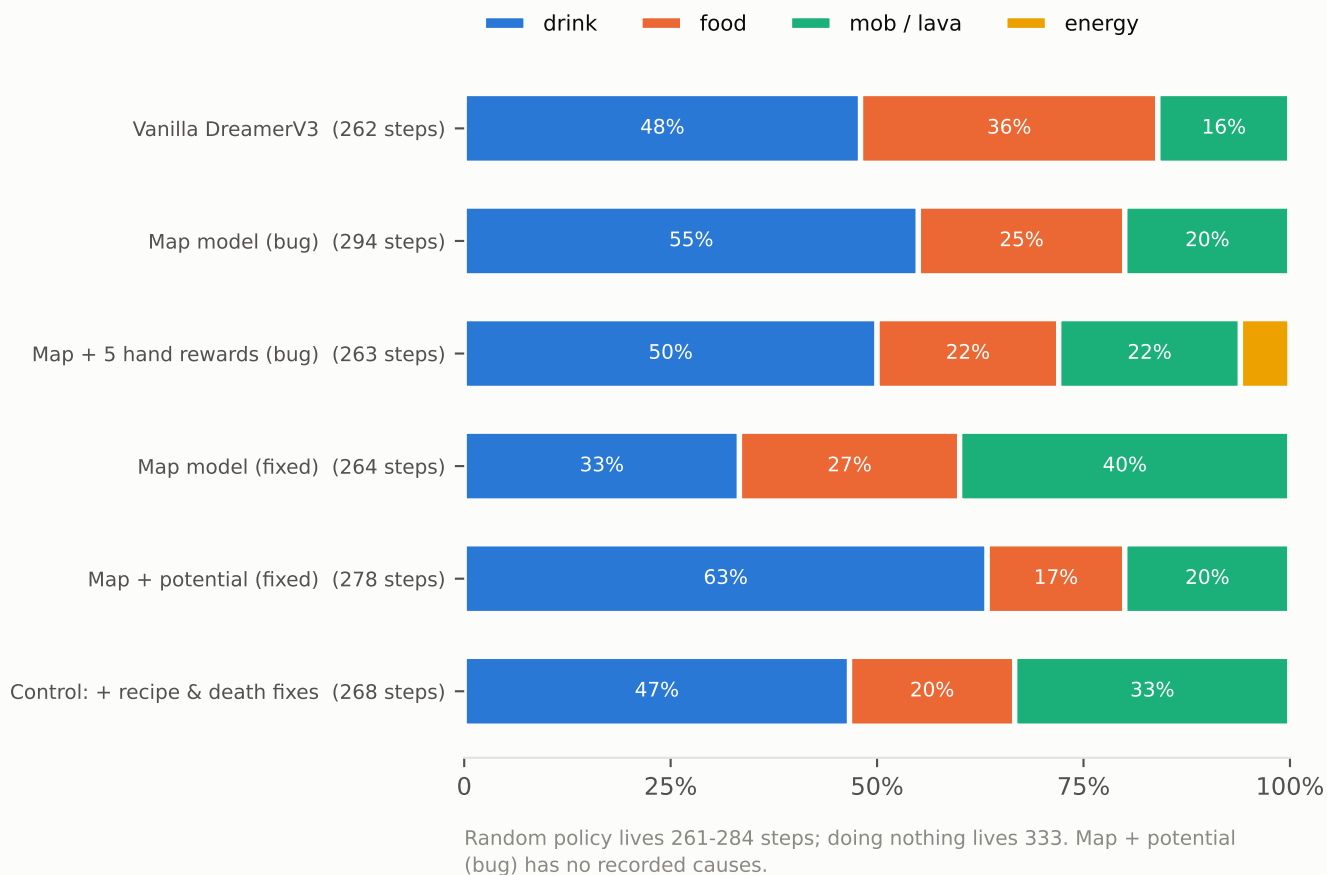
Every achievement any run unlocked

% of evaluation episodes in which each achievement fired. The other 54 of 67 were never unlocked by any run.

	vanilla	map, bug	map+5 terms	map+ Φ , bug	map	map+ Φ	map+ Φ +fixes
collect wood	76	72	52	63	83	100	97
place table	64	35	.	13	63	90	83
eat cow	4	12	22	13	10	3	17
collect sapling	92	92	98	90	93	93	93
collect drink	80	43	44	37	70	60	67
make wood pickaxe	24	.	.	.	7	30	.
make wood sword	.	.	.	.	.	10	60
place plant	92	77	.	73	93	87	83
defeat zombie	.	5	12	3	.	.	3
collect stone	12	.	.	.	.	20	.
place stone	.	.	.	.	.	3	.
defeat skeleton	.	2	2	.	.	.	.
wake up	.	98	.	93	83	83	90

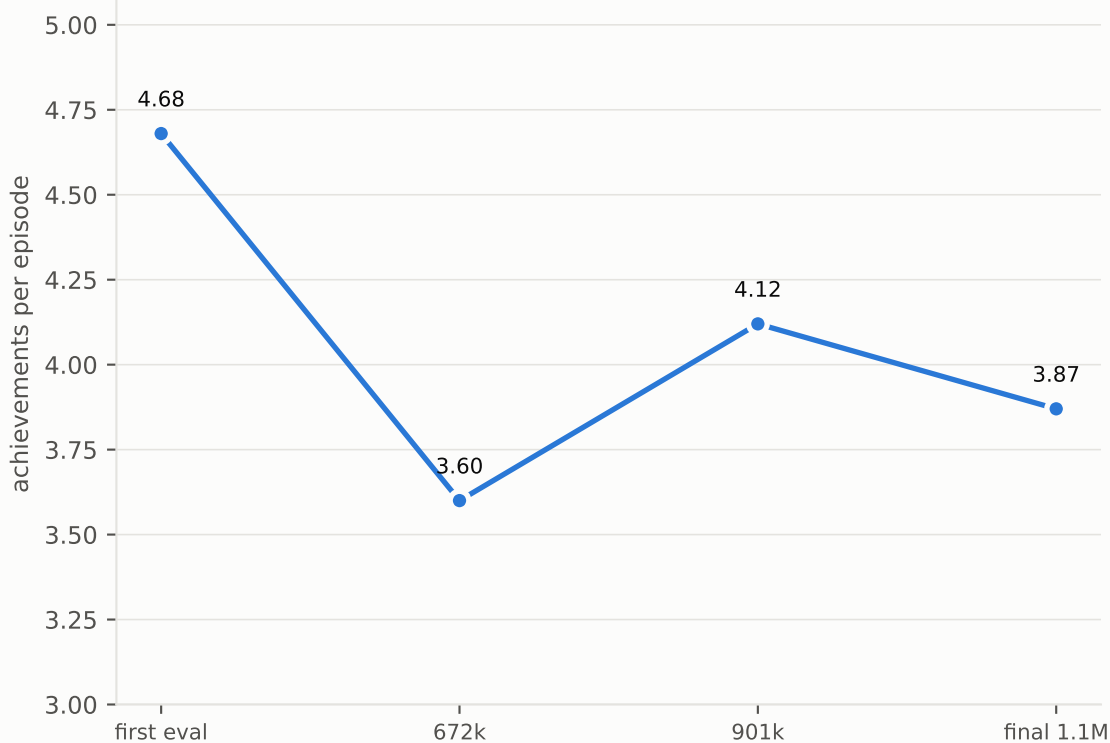
Survival never moved

Cause of death (share of evaluation episodes) and lifespan; and how much one run moves between its own checkpoints



One run, four checkpoints: Map + potential (bug)

The same training run, scored at four points (25-30 episodes each), moves by 1.08 achievements. That is the scale of noise any comparison between single runs has to beat.



Why the newest result is suspicious, and what would settle it

1. The pickaxe disappeared

Old record: pickaxe 30%, sword 10%, table 90%. Control with the recipe and death fixes: pickaxe 0%, sword 60%, table 83%. Both crafts need the same thing: one wood, standing next to a table. The agent reaches that state in most episodes, and it now consistently presses one crafting key and never the other. The potential weights the pickaxe three times the sword (3.0 vs 1.0), so the shaping pushes the other way. Across all seven runs only three ever made a pickaxe (vanilla 24%, map 7%, old record 30%) and only two ever made a sword.

2. The map model's gain is mostly sleep

WAKE_UP fires in 83-98% of episodes for every map run that learned to sleep, and 0% for vanilla. Remove it and the fixed map model (4.20) is below vanilla (4.44); map + potential (4.97) and the control (5.03) stay above it. So the evidence that the map itself helps is weak; the potential is what adds tech progress. Why map runs learn to sleep is an open question. It is systematic (every map run except the hand-written rewards one does it), not seed noise.

3. It is not the measurement

Every run was scored on the same fixed worlds, and re-scoring two old checkpoints with today's code reproduced them to the last digit. The pickaxe gap (9/30 vs 0/30) is far larger than evaluation noise.

4. Three explanations for the pickaxe, and the test for each

(a) Run-to-run variance. One training seed per run, and one run moves 3.60-4.68 between its own checkpoints. Two trainings of identical code can settle into different habits. Test: a second seed of the control configuration.

(b) Valid-action suppression (Zabounidis et al. 2026, verified). Crafting keys do nothing in most states, so the gradient pushes them down there, and shared weights carry that into the table states where they work. Which key survives may be close to arbitrary per run, which fits (a) too. Test: experiment A, the probability of each crafting key in table-with-wood states for both checkpoints; no training needed.

(c) The fixes changed the incentive. Before the death fix, dying kept the potential, including the pickaxe's capability term (2.0 per tier held), so a pickaxe held at death was a permanent bonus. The old 30% may have been partly propped up by that bug. Test: honest_map carries the same fixes; if it also skips the pickaxe, the fixes are implicated, and a run with only the death fix would isolate it.

5. What to run, cheapest first

Experiment A on the control and the old record (hours, no training). Score honest_map when it finishes. Then second seeds of the control and the old record, since single runs cannot separate (a) from (c), and reviewers expect 5+ seeds anyway. Explain the WAKE_UP gap before claiming the map model helps.